

Q1(a) Qualification document, Test lists and Procedures

1. Visual & Pre-Power Checks

In this test we need to check PCB inspection, Solder joints Quality, Component orientation and Connector integrity.

Procedure: -

1. Inspect under microscope
2. Check against schematic & BOM

2. Voltage Accuracy Test

In this test whether all the output voltages are within the allowed limit of tolerance will be checked. So, let's suppose if we fix the tolerance as $\pm 5\%$ then we need to check all the output against tolerance value

Procedure: -

1. Apply +5 V input
2. Measure output using DMM
3. Compare with limits

3. Load Regulation

In this test whether the output voltage will be remain stable under the load will be tested

Procedure: -

1. Apply min to max load using electronic load
2. Measure voltage change

4. Ripple & Noise Test

In this test whether the ripple and noise will make damage to sensitive payload electronics will be tested

As specified in the document Ripple will be tested in the condition that **Ripple < 50 mV p-p**

Procedure: -

1. Measure ripple using oscilloscope

2. Use short ground spring
3. Measure under max load

5. Line Regulation

In this test PDN will be tested with small input variations to test PDN tolerance

Procedure: -

1. Vary input voltage from 4.75V to 5.25V and observe the output stability during the input voltage Variation

6. Power-Up / Power-Down Sequencing

In this test sequence of power up and power down will be tested to ensure Incorrect sequencing damaging which may cause damage to the ICs

Procedure: -

1. Monitor rails simultaneously on scope
2. Verify order matches design
3. Confirm PGood behavior

7. Load Transient Test

In this test sudden load change response will be checked

Procedure: -

1. Apply fast load step from 10% to 90% and Measure undershoot, overshoot, settling time

8. Protection Tests

In this test Safety & robustness will be checked by conducting Over-current, short circuit, Thermal protection Testes

Q1(b) Problem: - Measured ripple = **120 mV p-p**, Specification = **< 50 mV p-p** and Issue seen **on all rails**

So, the issue seen in all the rails so something fundamental is wrong not in single rail

In this case Insufficient output capacitance, Incorrect capacitor type, Poor PCB layout, Measurement error, switching instability, Input noise coupling will be checked to solve the problem.

Q1(c) Design improvement recommendations

1. Testability Improvements:- Dedicated Test points will be provided at each voltage rails, ground and PGood signals points.
2. Production Improvements:- Using of common capacitor values, avoid tight tolerance, add footprint options for capacitors, Clear silk labels for rails Can be done
3. Reliability Improvements: - More margin on current rating, Thermal vias under regulators, Bulk and ceramic capacitor combination by this way we can increase product reliability

Q1(d): Production Acceptance Tests for 50 units

So, the flow will be as below

1. Incoming Inspection

- Visual
- Component verification

2. Pre-Power Checks

- Shorts between rails & ground

3. Power-Up Test

- Apply +5 V
- Check startup current

4. Voltage Test

- All rails measured

5. Ripple Test

- Spot-check or sample

6. Load Test

- Nominal load on each rail

7. Power Sequencing

- Sampling basis

8. Automated Report

- Pass / Fail
- Serial number logged

Q2(a) Detail explanation and code will be present in the GitHub and its link will be <https://github.com/Malegowda1/Python-script-for-load-transient-test>

Q2(b) Failure Analysis Report — Board SN-017

1. Failure Summary

Board SN-017 passes all automated DC voltage and ripple tests but exhibits intermittent failures during the load transient test on the 3.3 V rail. Out of 10 repeated captures, 3 captures failed to meet the specified transient limits while 7 remained within specification.

2. Observations

- Failure is limited to the 3.3 V rail
- DC voltage and steady-state ripple remain within specification
- Failure is intermittent and non-deterministic
- Indicates marginal transient performance rather than a hard fault

3. Failure Analysis Approach

The failure analysis focused on confirming the validity of the measurement, comparing pass and fail waveforms, evaluating load step conditions, checking temperature sensitivity, and inspecting rail-specific components. Additional emphasis was placed on identifying stability margin issues and component tolerances affecting transient response.

4. Root Cause Assessment

Based on waveform behaviour and repeatability, the failure is most likely caused by marginal loop stability or output capacitance sensitivity on the 3.3 V rail. Component tolerance, MLCC DC bias derating, or assembly-related variations may contribute to the intermittent behaviour.

5. Recommendation

Board SN-017 should not be released for flight or production use in its current state. Intermittent transient failures present unacceptable reliability risk. The board should be classified as suspect, subjected to corrective action or rework, and the design and production test coverage should be updated to ensure adequate stability margin across all units.

